

الورقة الامتحانية
لامتحان الشهادة الثانوية الأزهريّة - نموذج تجريبي
للعام الدراسي ١٤٤٧ هـ - ٢٠٢٥ / ٢٠٢٦ م

المادة/الأحياء (لغات) (تجريبي)	
القسم العلمي	زمن الإجابة: ثلاث ساعات

السؤال	الدرجة بالأرقام	اسم المصحح ثلاثياً
الأول		
الثاني		
الثالث		
الرابع		
الخامس		
المجموع		

عدد الصفحات (١٦) صفحة
و على الطالب مسئولية المراجعة
والتأكد من ذلك قبل تسليم الكراسة

الرقم السري

مجموع الدرجات (مكتوباً بالحروف):

اسم المراجع العددي ثلاثياً: اسم المراجع الفني ثلاثياً:

الأزهر الشريف - قطاع المعاهد الأزهريّة - الإدارة المركزيّة للامتحانات وشؤون الطلاب و الخريجين

عدد الصفحات (١٦) صفحة
و على الطالب مسئولية المراجعة
والتأكد من ذلك قبل تسليم الكراسة

امتحان الشهادة الثانوية الأزهريّة - تجريبي

للعام الدراسي ١٤٤٧ هـ - ٢٠٢٥ / ٢٠٢٦ م

المادة/الأحياء (لغات) (تجريبي)

الرقم السري

نموذج ثانوية أزهريّة تجريبي

اسم الطالب كاملاً:

اسم المعهد:

رقم الجلوس:

التاريخ: / /

تعليمات هامة

عزيزي الطالب / عزيزتي الطالبة:

- اقرأ السؤال بعناية، وفكر فيه جيداً قبل البدء في الإجابة عليه.
 - أجب عن جميع الأسئلة ولا تترك أي سؤال دون إجابة.
 - عند إجابتك عن الأسئلة المقالية، أجب فيما لا يزيد عن المساحة المحددة لكل سؤال.
 - عند إجابتك عن أسئلة الاختيار من متعدد (إن وجدت)، ظلل الدائرة ذات الرمز الدال على الإجابة الصحيحة فقط تظليلاً كاملاً.
- مثال: الإجابة الصحيحة (د) مثلاً

أ	ب	ج	د
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- في حالة التظليل على أكثر من رمز، تعتبر الإجابة خطأ.
 - في حالة ما إذا أجبنا إجابة خطأ، ثم قمنا بشطبها وأجبنا إجابة صحيحة تُحسب الإجابة صحيحة.
 - في حالة ما إذا أجبنا إجابة صحيحة، ثم قمنا بشطبها وأجبنا إجابة خطأ تُحسب الإجابة خطأ.
- ملحوظة:** يفضل عدم تكرار الإجابة على الأسئلة.
- عدد صفحات الكراسة (١٦) صفحة.
 - تأكد من عدد صفحات كراستك، فهي مسئوليتك.
 - زمن الامتحان (ثلاث ساعات).
 - الدرجة الكلية للامتحان (٦٠) درجة.
 - عدد الأسئلة (٥) لكل سؤال (١٢) درجة ، لكل جزئية (درجة واحدة)
 - عند احتياج الطالب للإجابة على أي فقرة وذلك عند حدوث أي سبب يقتضي ذلك؛ يستخدم المسودة بآخر الورقة الامتحانية مع كتابة رقم السؤال والفقرة بوضوح، بشرط ألا تكون الإجابة مكررة.

هذا الجزء

غير مخصص للإجابة

Answer the following questions:

(12 Marks for each)

[First question]

(A) Shade the circle with the symbol denoting the correct one for each of the following:

1- A group of hormones responsible for the growth of the prostate gland and seminal vesicles in males

- | | |
|---------------|----------------|
| (a) Androgens | (b) Oestrogens |
| (c) TSH | (d) ACTH |

2- Which of the following living organisms reproduces by regeneration?

- | | |
|----------------|------------|
| (a) Shrimp | (b) Frog |
| (c) Salamander | (d) Sponge |

3- Joints are found between the centra of the vertebrae of the vertebral column of human.

- | | |
|----------------------------|-------------------------------|
| (a) Fibrous | (b) Synovial limited movement |
| (c) Synovial wide movement | (d) Cartilaginous |

4- The substance that stimulates the maturation of lymphoid stem cells into T-cells is:

- | | |
|----------------|------------------|
| (a) Chemokines | (b) Interleukins |
| (c) Thymosin | (d) Interferons |

5- To form a protein of 300 amino acids, the number of turns of DNA molecule which is replicated to form this protein is turns.

- | | |
|--------|--------|
| (a) 15 | (b) 30 |
| (c) 45 | (d) 90 |

6- molecule is responsible for both the translation of amino acids and nucleotides.

- | | |
|--------------------|----------|
| (a) RNA polymerase | (b) tRNA |
| (c) mRNA | (d) DNA |

(B) Write the scientific term for each of the following statements:

1- The enzyme that synthesizes DNA using mRNA as a template.

(.....)

2- A defense system depending on successive non-specific mechanisms that surround invading microbes to prevent their spread.

(.....)

3- The nutritive tissue surrounding the embryo sac inside the flower ovary.

(.....)

(C) Give reason for:

1- Sometimes spirogyra performs conjugation.

.....

2- Skeletal muscle fatigue occurs.

.....

3- About 5000 purine bases are lost daily from the DNA of each human cell.

.....

[Second question]

(A) Shade the circle with the symbol denoting the correct one for each of the following:

- 1- The cloning of Dolly ewe was from

(a) zygote nucleus	(b) mother ova
(c) Cells from mother's breast	(d) zygote cells

- 2- Number of types of cells in the seminiferous tubules are

(a) 2	(b) 4
(c) 6	(d) 8

- 3- Complements couldn't perform its role except in the presence of cells.

(a) T _C	(b) T _S
(c) T _H	(d) B

- 4- Increased excitability and angry in man is due to hormone.

(a) decrease of adrenaline	(b) increase of adrenaline
(c) decrease of parathormone	(d) increase of thyroxin

- 5- In the m-RNA piece which is replicated from one of DNA strand :
5' - GUC CCA GCU CCA GCU UGA -- 3'
the number of types of t-RNA molecules used to translate this part is

(a) 6	(b) 5
(c) 4	(d) 3

- 6- The plant surrounds itself by a layer of impermeable cells and thickened with substance.

(a) Cutin	(b) Cellulose
(c) lignin	(d) suberin

(B) Study the opposite figure which represents one of steps of protein synthesis, and answer the following:

1- This step indicates

- (a) start of protein synthesis (b) polypeptide chain elongation
 (c) stop of protein synthesis (d) building of mRNA

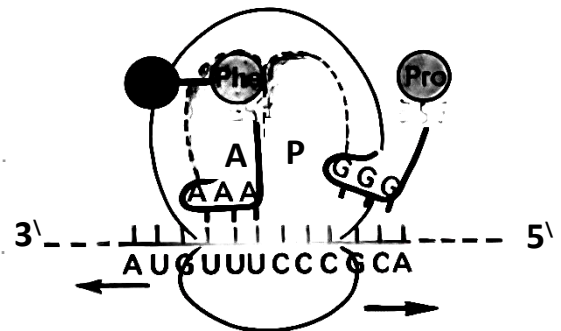
2- Find the mistakes in this figure & write the Correction:

- The mistake **(1)** is:

- The correction:

- The mistake **(2)** is:

- The correction:



3- Determine the functional structure of this organelle:

a)

b)

C- Correct the following statements without changing the underlined words:

1. The female organ in the life cycle of Polypodium is the antheridium.

2. Number of bones of the pectoral girdle that attached to the sternum is 20.

3. Insulin hormone is secreted from Alpha cells of islets of Langerhans.

4. The site of connection of Ribosomes with mRNA molecule is CCA.

[Third question]

(A) Shade the circle with the symbol denoting the correct one for each of the following:

1- gland is connected with the hypothalamus.

- | | |
|---------------|-----------------|
| (a) Thymus | (b) Thyroid |
| (c) Pituitary | (d) Parathyroid |

2- The number of Free phosphate groups in the plasmid is

- | | |
|-------|----------|
| (a) 3 | (b) 2 |
| (c) 1 | (d) Zero |

3- Rhizome is found in

- | | |
|-------------------------------|------------------------------|
| (a) gametophyte of Polypodium | (b) bread mould fungus |
| (c) bean plant | (d) sporophyte of Polypodium |

4- Immune cells are difficult to recognize patient.

- | | |
|---------------|-----------------|
| (a) influenza | (b) Corona |
| (c) AIDS | (d) chicken pox |

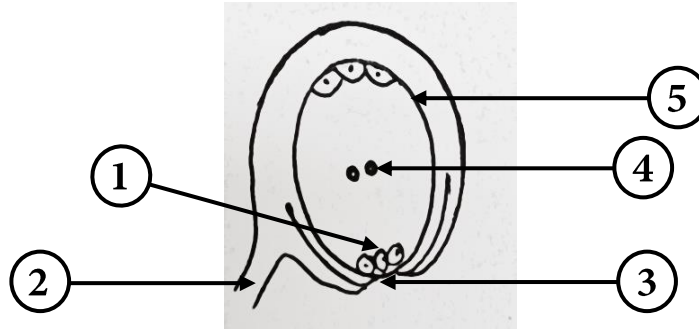
5- Number of Ligaments that join femur with tibia

- | | |
|-------|-------|
| (a) 1 | (b) 2 |
| (c) 3 | (d) 4 |

6- In the sperm, the two centrioles and acrosome are present respectively in:

- | | |
|--------------------|--------------------|
| (a) head, neck | (b) neck, head |
| (c) midpiece, head | (d) head, midpiece |

(B) 1) Study the opposite figure, which shows a schematic diagram of an ovule just before fertilization, then complete the table.



Indication / Function	Structure Number	Name of Structure
Female gamete		
Forms the endosperm		
Through which the pollen tube enters		
Connects the ovule to the ovary wall		

2- What will happen when the contents of pollen tube transfer to inside the structure (5).

.....

.....

(C) Mention:

1- The way of asexual reproduction in tobacco plant.

(.....)

2- A genetic case in human that contains 45 chromosomes in its somatic cells.

(.....)

3- Hormone stimulates the contraction of the gall bladder.

(.....)

[Fourth question]

(A) Shade the circle with the symbol denoting the correct one for each of the following:

1- Helper T cells can't recognize Antigens except after its binding to

- | | |
|-------|--------|
| Ⓐ Ig | Ⓑ MHC |
| Ⓒ CD8 | Ⓓ ACTH |

2- Nucleic acids RNA consist of building units called

- | | |
|-------------------|---------------|
| Ⓐ genes | Ⓑ Nucleotides |
| Ⓒ Ribonucleotides | Ⓓ (a) and (c) |

3- Which of the following is not from the components of pollen grains after its releasing from the anther

- | | |
|----------------------|---------------|
| Ⓐ tube nucleus | Ⓑ pollen tube |
| Ⓒ generative nucleus | Ⓓ thick wall |

4- The mutation in the penicillium is

- | | |
|----------------------|-----------------------|
| Ⓐ spontaneous useful | Ⓑ spontaneous harmful |
| Ⓒ induced useful | Ⓓ induced harmful |

5- If the number of chromosomes in the nucleus of pea plant cell equals 7 pairs, then the number of chromosomes in the tube nucleus is:

- | | |
|-----------------|------------|
| Ⓐ 7 chromosomes | Ⓑ 7 pairs |
| Ⓒ 21 chromosome | Ⓓ 14 pairs |

6- Heart muscle is

- | | |
|------------------------|--------------------------|
| Ⓐ voluntary striated | Ⓑ involuntary unstriated |
| Ⓒ involuntary striated | Ⓓ involuntary unstriated |

(B) Choose from column B, C which is suitable for column (A) to form correct statements.

N	(A) Substance	(B) Type of secreted cells	(C) Importance
1-	Histamine	a- Activated TC	a- activate lymphocytes
2-	Perforin	b- Activated TS	b- Stop the activation of Lymphocytes
3-	Lymphokines	c- Mast cell	c- destroy cancer cells
4-	Cytokines	d- Activated TH	d- prevent the spread of virus in the healthy cell
5-	Interferons	e- Cancer cells	e- increase the permeability of blood capillaries to blood fluids

- 1)
- 2)
- 3)
- 4)
- 5)

(C) Show by drawing the role of enzymes in DNA replication.

[Fifth question]

(A) Shade the circle with the symbol denoting the correct one for each of the following:

1- Breccia's (sharp edges) is from detritus sedimentary rocks, that its volume of granules is

- (a) greater than 2mm (b) from 2mm - 62 micron
(c) from 62 - 4 micron (d) less than 4 micron

2- The average of thickness of the Earth's crust is

- (a) 8 Km (b) 10 Km
(c) 34 Km (d) 60 Km

3- is from the metamorphic rocks, its structure is SiO_2 .

- (a) sand stone (b) quartzite
(c) Marble (d) limestone

4- A branch of geology which deals with the different structures which exist on rocks resulting from the effect of both external and internal forces is

- (a) physical (b) structural
(c) engineering (d) stratigraphy

5- In the opposite crystal system, the length of its crystal axis is

- (a) $a_1 = a_2 = a_3$ (b) $a_1 = a_2 \neq c$
(c) $a \neq b \neq c$ (d) $a_1 = a_2 = a_3 \neq c$

6- The chemical structure of Malachite mineral is

- (a) zinc sulphide
(b) hydrous copper carbonate
(c) hydrous calcium sulphate
(d) Anhydrous calcium sulphate.

(B) Study the opposite figure & answer the following:

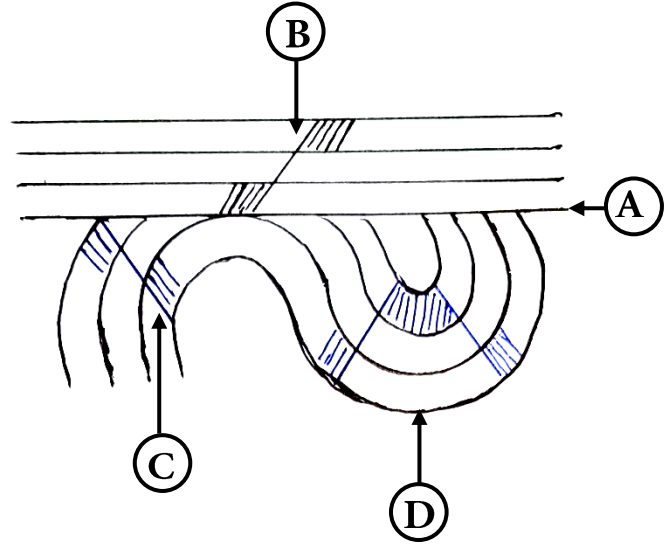
1- What is the type of unconformities in this figure?

2- What is the mass (d):

3- The number of types of tectonic structures in this figure is:

- (a) 2 (b) 4
(c) 6 (d) 8

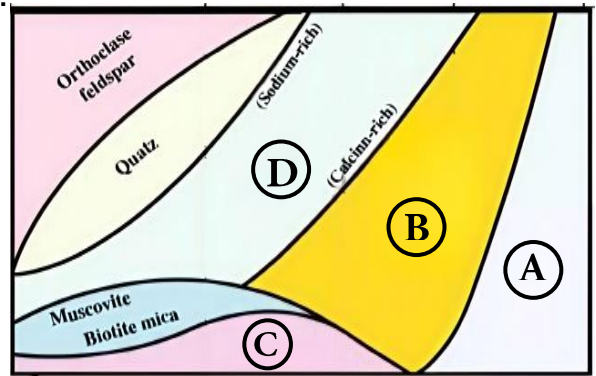
4- Mention the similarity between structure (b) and (c).



(C) The opposite figure explains the structure of common Igneous rocks answer the following questions:

1) determine what these letters refers to:

- a)
b)
c)
d)



2) Give one example for:

- a) Plutonic Igneous rock rich with mineral structure (a).
.....
b) Plutonic Igneous rock, which quartz represents 1/4 of its volume.
.....
c) Extruded (Volcanic) igneous rock in which silica contains 55-66%.
.....

